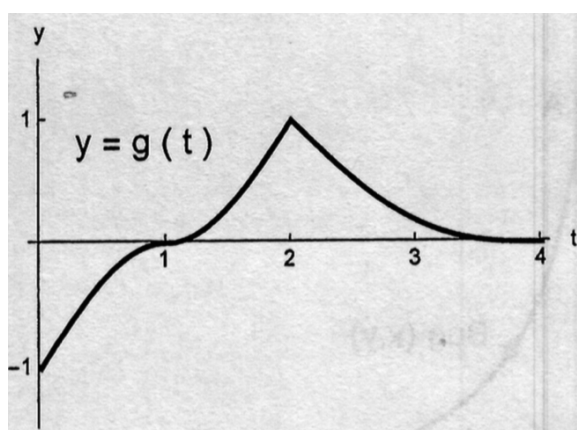


Problem 7

Problem 7

Problem 1 The graph of g , a continuous function $[0, 4]$, is shown in the figure. Let $A(x) = \int_0^x g(t) dt$ for $0 \leq x \leq 4$.



(a) Circle the correct statement about $A(1)$.

- (i) $A(1) = 0$;
- (ii) $A(1) < 0$;
- (iii) $A(1) > 0$;
- (iv) none of the previous answers.

Explanation. Correct choice: (ii) $A(1) < 0$. $A(1)$ is the net area of the region bounded by the curve and the x -axis between $x = 0$ and $x = 1$. On this interval, $g(t)$ is negative and therefore, $A(1)$ is negative.

(b) Circle the correct statement about $A(1.5)$.

- (i) $A(1.5) = 0$;
- (ii) $A(1.5) < 0$;
- (iii) $A(1.5) > 0$;
- (iv) none of the previous answers.

Explanation. Correct choice: (ii) $A(1.5) < 0$. The net area on the interval $(1, 1.5)$ is positive and smaller than the negative net area on the interval $(0, 1)$. Therefore, the net area on the interval $(0, 1.5)$ is negative.

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(c) Circle the correct statement about $A'(1.5)$.

- (i) $A'(1.5) = 0$;
- (ii) $A'(1.5) < 0$;
- (iii) $A'(1.5) > 0$;
- (iv) none of the previous answers.

Explanation. Correct choice: (iii) $A'(1.5) > 0$ because $A'(1.5) = g(1.5)$

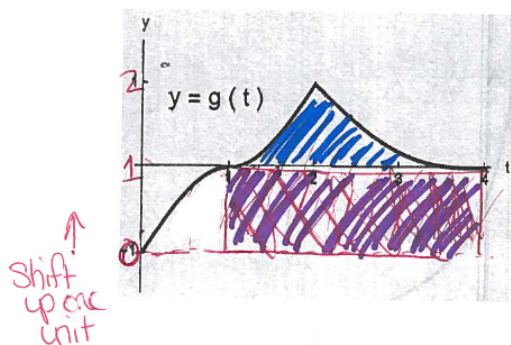
(d) Circle the correct expression for $\int_1^4 (g(t) + 1) dt$.

- (i) $A(4) + 1$;
- (ii) $A(4) - A(1)$;
- (iii) $A(4) - A(1) + 1$;
- (iv) $A(4) + 3$;
- (v) $A(4) - A(1) + 3$;
- (vi) none of the previous answers.

Explanation. Correct choice: (v) $A(4) - A(1) + 3$

We can split the integral: $\int_1^4 (g(t) + 1) dt = \int_1^4 g(t) dt + \int_1^4 1 dt = A(4) - A(1) + 3$

We can visualize this integral by imagining shifting the graph of g up one unit. See figure below. Our area we are looking for consists of the area under the curve g on the interval $(1, 4)$ (the blue region) plus the rectangular region formed below (the purple region).



The area of the blue region is $\int_1^4 g(t) dt = \int_0^4 g(t) dt - \int_0^1 g(t) dt = A(4) - A(1)$.

The area of the purple region is $\int_1^4 1 \, dt$. This integral is the area under the line $y = 1$ on the interval $(1, 4)$ which is a rectangular region of area 3.

Combining these, the total area $\int_1^4 (g(t) + 1) \, dt = A(4) - A(1) + 3$

(e) Circle the correct statement about $A(0)$.

- (i) $A(0) = 0$;
- (ii) $A(0) = 1$;
- (iii) $0 < A(0) < 1$;
- (iv) $A(0) > 1$;
- (v) $A(0) < 0$;
- (vi) none of the previous answers.

Explanation. Correct choice: (i) $A(0) = 0$. By the definition of A , $A(x) = \int_0^x g(t) \, dt$ for $0 \leq x \leq 4$. So, $A(0) = \int_0^0 g(t) \, dt = 0$.

(f) Circle the interval (or intervals) where the function A is DECREASING.

- (i) $(0, 1)$;
- (ii) $(1, 2)$;
- (iii) $(2, 4)$;
- (iv) none of the previous answers.

Explanation. Correct choice: (i) $(0, 1)$. A is decreasing means that the area is decreasing or we are accumulating negative area. This happens when A 's derivative, g is negative.

(g) Circle the value (or values) where the function A attains its MAXIMUM.

- (i) $x = 0$;
- (ii) $x = 1$;
- (iii) $x = 2$;
- (iv) $x = 3$;
- (v) $x = 4$;
- (vi) none of the previous answers.

Explanation. Correct choice: (v) $x = 4$. It appears that $A(x) < 0$ on $(0, 2)$ and $A(x) > 0$ on $(2, 4]$. So, the maximum occurs on $[2, 4]$. Since $A' = g$ is positive on $(2, 4)$, and A is increasing there, the maximum is attained at the end point, $x = 4$. In other words, A achieves a maximum when we have the largest positive area. Because we start with negative area and then continue to accumulate positive area until $x = 4$, the maximum occurs at the end point.

(h) Circle the value (or values) where the function A attains its MINIMUM.

- (i) $x = 0$;
- (ii) $x = 1$;
- (iii) $x = 2$;
- (iv) $x = 3$;
- (v) $x = 4$;
- (vi) none of the previous answers.

Explanation. Correct choice: (ii) $x = 1$. $A' = g$ is negative on $(0, 1)$ and positive on $(1, 4)$. So A is decreasing on $(0, 1)$ and increasing on $(1, 4)$. So the local and absolute maximum occurs at $x = 1$. In other words, A achieves a minimum when we have the largest negative area. We start by accumulating negative area and then switch to accumulating positive area at $x = 1$.

(i) Circle the interval (or intervals) where the function A is CONCAVE DOWN.

- (i) $(0, 1)$;
- (ii) $(1, 2)$;
- (iii) $(2, 4)$;
- (iv) none of the previous answers.

Explanation. Correct choice: (iii) $(2, 4)$. A is concave down where $A' = g$ is decreasing. g is decreasing on $(2, 4)$.

(j) Circle the value (or values) of x where the function A has an inflection point.

- (i) $x = 1$;
- (ii) $x = 2$;
- (iii) $x = 3$;
- (iv) none of the previous answers.

Explanation. Correct choice: (ii) $x = 2$. $A' = g$ changes from increasing to decreasing.

(k) Sketch the graph of A , based on items (e)-(j)

Explanation.

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